CHAPTER 2

COMPONENTS OF THE COMPUTER SYSTEM

CHAPTER ESSENCE AT A GLANCE:
Computer systems unite tangible electronic hardware with system and application software through system buses and operating system supervisors to coordinate resource scheduling.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Hardware and Software Symbiosis:** Hardware represents the physical electronic circuitry and mechanical components of a computer, whereas software encompasses the structured sequences of instructions that guide hardware execution.

• **System Bus Triad and Signal Segregation:** The system bus comprises the bidirectional data bus for transferring values, the unidirectional address bus for locating memory cells, and the control bus for synchronizing read and write command pulses.

• **Operating System as Resource Supervisor:** The operating system acts as an extended machine and master resource manager, overseeing processor scheduling, memory allocation, device communication, and file security.

• **Language Processors and Open Source Paradigm:** Translators bridge human-readable programming languages to raw machine code through compilers and interpreters, while Free and Open Source Software (FOSS) guarantees collective software freedoms.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{System Bus} = \text{Data Bus (Bidirectional)} + \text{Address Bus (Unidirectional)} + \text{Control Bus}$$

Three-Tier System Bus Functional Composition

$$\text{Addressable Locations} = 2^{k} \quad (k = \text{Address Bus Width in Bits})$$

Direct Memory Address Space Formula

$$\text{Source Code} \xrightarrow{\text{Compiler or Interpreter}} \text{Object Code} \xrightarrow{\text{Linker and Loader}} \text{Executable In-Memory}$$

Language Processor Translation Pipeline

$$\text{Software Freedom} = \text{Run (F0)} + \text{Study (F1)} + \text{Redistribute (F2)} + \text{Improve (F3)}$$

FSF Four Essential Software Freedoms

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The address bus is strictly unidirectional originating from the CPU, while the data bus is bidirectional allowing reading and writing.

• **Alert:** Compilers translate the entire source program into machine code in a single batch, whereas interpreters translate and execute line by line.

• **Alert:** Do not confuse freeware with open source software; freeware is free of monetary charge but proprietary, while open source grants access to modify source code.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is a system bus? Differentiate between the data bus and address bus. [Score: 2 Marks]

Answer & Key Valuation Steps:

• System Bus Definition: A set of shared parallel electrical conductors that transmit electrical signals and facilitate communication among CPU, memory, and peripheral controllers. **[0.5 Mark]**

• Data Bus: Bidirectional bus carrying data operands and instructions between the CPU and memory or input and output ports. **[0.75 Mark]**

• Address Bus: Unidirectional bus originating from the CPU carrying binary memory addresses to pinpoint specific memory cells. **[0.75 Mark]**

2. Distinguish between system software and application software with illustrative examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• System Software: Programs designed to manage, control, and extend the computational capabilities of the computer hardware (e.g. Operating Systems like Linux, Windows, Compilers). **[1 Mark]**

• Application Software: Software created to accomplish specific end-user business or academic tasks (e.g. Word processors, Spreadsheets, Web browsers). **[1 Mark]**

3. Compare compilers and interpreters with respect to translation approach, speed, and error reporting. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Translation Approach: A compiler scans the entire high-level source code file at once and generates an independent object code file; an interpreter translates and executes source code line by line without creating an object file. **[1 Mark]**

• Execution Speed: Compiled machine code executes at maximum hardware speed during runtime; interpreted code executes slower because translation happens continuously during execution. **[1 Mark]**

• Error Diagnostics: Compilers list all syntax errors collectively after full compilation; interpreters stop execution immediately upon encountering the first erroneous line, facilitating interactive debugging. **[1 Mark]**

4. Explain the four core resource management functions of an Operating System. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Process Management: Allocates CPU time to active processes using scheduling algorithms and provides synchronization. **[0.75 Mark]**

• Memory Management: Tracks memory usage, allocates memory blocks to running applications, and reclaims freed blocks. **[0.75 Mark]**

• File Management: Organizes data files into structured hierarchical directories, managing storage access permissions and security. **[0.75 Mark]**

• Device Management: Controls communication with peripheral devices using specialized software drivers and input-output buffers. **[0.75 Mark]**

5. Explain cache memory and describe how it overcomes the speed disparity between CPU and main memory. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Cache Memory Architecture: High-speed static RAM (SRAM) memory positioned between the CPU and dynamic RAM (DRAM) main memory to store frequently referenced instructions and data. **[1 Mark]**

• Principle of Locality of Reference: Takes advantage of temporal locality (recently accessed items are likely to be accessed again soon) and spatial locality (nearby memory items will likely be requested soon). **[1 Mark]**

• Performance Impact: Cache access takes between 2 and 10 ns compared to 50 ns for main memory, substantially reducing CPU idle wait states and boosting overall execution throughput. **[1 Mark]**

6. Describe the four essential software freedoms defined by the Free Software Foundation (FSF). [Score: 3 Marks]

Answer & Key Valuation Steps:

• Freedom 0: The freedom to run the program for any purpose without commercial or personal restriction. **[0.75 Mark]**

• Freedom 1: The freedom to study how the program works and modify its source code to suit specific needs (access to source code is a prerequisite). **[0.75 Mark]**

• Freedom 2: The freedom to redistribute copies of the program to assist peers and community members. **[0.75 Mark]**

• Freedom 3: The freedom to improve the program and release your enhancements to the public so that the whole community benefits. **[0.75 Mark]**

7. Differentiate between proprietary software, freeware, shareware, and open source software. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Proprietary Software: Commercial software owned by an individual or corporation; source code is kept secret and users purchase a restrictive license to run it (e.g. Microsoft Windows). **[1 Mark]**

• Freeware: Copyrighted proprietary software distributed free of financial charge, but the source code remains closed and modification is forbidden (e.g. Adobe Acrobat Reader). **[1 Mark]**

• Shareware: Demonstration software distributed free of cost for a trial period, after which users must pay a license fee for continued access (e.g. WinRAR). **[1 Mark]**

• Open Source Software: Software whose human-readable source code is made publicly available under open licenses granting freedom to inspect, modify, and redistribute (e.g. Linux, Python). **[1 Mark]**